

THE RELATIONSHIP BETWEEN GENTRIFICATION AND CRIME IN AUSTIN,
TEXAS FROM 2003 TO 2013

DIRECTED RESEARCH

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Presented to the Graduate College of Texas State University

In Partial Fulfillment of

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For the Degree of

Master of Applied Geography

By

Henry Reissig

San Marcos, Texas

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Committee Members Approved:

Name of Committee Chair, Chair

Directed Research Committee Member

Date

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By

Henry Reissig

2015

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Abstract

This study analyzes the relationship between gentrification and crime in Austin, Texas for U.S. census tracts determined to be gentrified between 2003 and 2013. With 24 gentrified and 24 non-gentrified tracts in the study, crime was split into three groups: property crime, violent crime, non-indexed crime. For each type of crime, the change in crime per 1000 people between the years 2003 and 2013 was calculated for gentrified, non-gentrified tracts along with the rest of Austin. The results show no statistically significant difference in the change in crime between gentrified and non-gentrified tracts. However, this study also divided all 48 gentrified and non-gentrified tracts into six parts based on location in the city, and this analysis revealed a significant difference in the average change in crime between gentrified and non-gentrified tracts based on where the tracts are located in the city. Overall, the results counter the perception that gentrified tracts are somehow safer than non-gentrified tracts and it might not be a valid reason to justify activities by government entities or investors aiming to develop economically depressed neighborhoods at risk for gentrification and help them make smarter choices in the planning to prevent this from happening.

I. INTRODUCTION

Gentrification of Austin

The Austin, Texas Metropolitan Area (MSA) population more than doubled between the years 1990 and 2010, from 846,227 to 1,716,289 people (Austin Chamber n.d.). Concern is growing over how this increase in population, and accompanying economic growth, have affected housing affordability in Austin. Specifically, the increase in property values and the subsequent increase in tax valuations in working class neighborhoods of Austin have driven a number of moderate to low income residents out of the area.

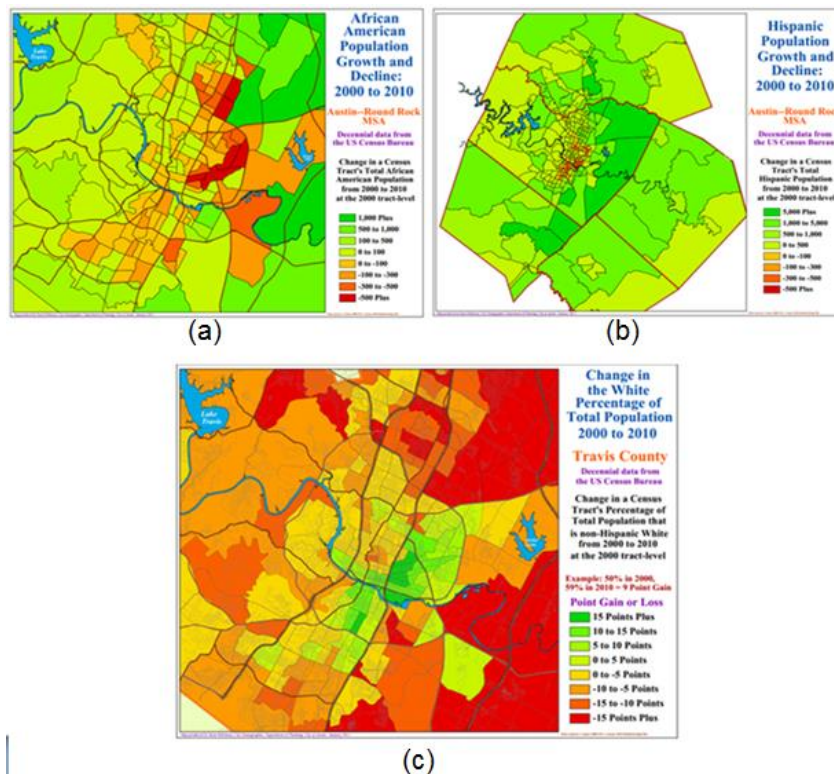


Figure 1. Race Migration Patterns

(Planning and Development-City of Austin Demographics Map Library 2015)

A higher percentage of African American and Hispanic households are more in the moderate to low income category than their white counterparts, making these groups more vulnerable. (Gentrification Committee Report 2001). This exodus is shown clearly in Figure 1 (a), (b) and (c). Figure 1(a) and Figure 1(b) show the change in census tract population for both the African American and Hispanic populations, respectively, and indicate a marked decrease for both populations in Austin's central corridor running the entire north-south length of the city. East central Austin, in particular, has seen a dramatic decrease in African American population and south central Austin has seen the same effect for the Hispanic population. Figure 1(c) shows the subsequent increase in white population for these same tracts. The number of building permits has also grown significantly since 1990, indicating a surge in new and remodeled homes (Figures 2-3). From this, it seems reasonable to claim that some degree of gentrification is taking place in Austin.

“Gentrification” is defined by the Brookings Institute as “the process by which higher income households displace lower income residents of a neighborhood, changing the essential character and flavor of that neighborhood” (Gentrification Committee Report 2001). Kirk and Laub 2010 give a way to think of gentrification as a reversal of the mass exodus from the central city from the past. They give three stages of neighborhood status: development, transition (downgrading, thinning out), and renewal/gentrification. In the development stage, single family homes are developed and increase the population density and the socioeconomic status of the neighborhood. In the transition stage, the population thins out for a variety of reasons including economic

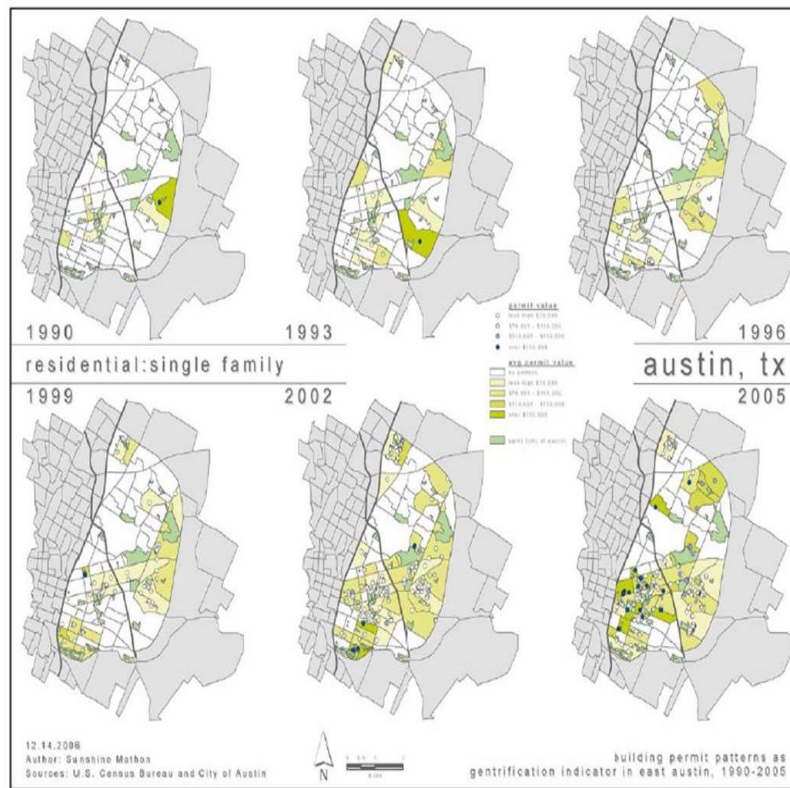


Figure 2. Increase in Building Permits for East Austin (1990-2005), (Single family residential and residential remodel)

(Mathon 2014)

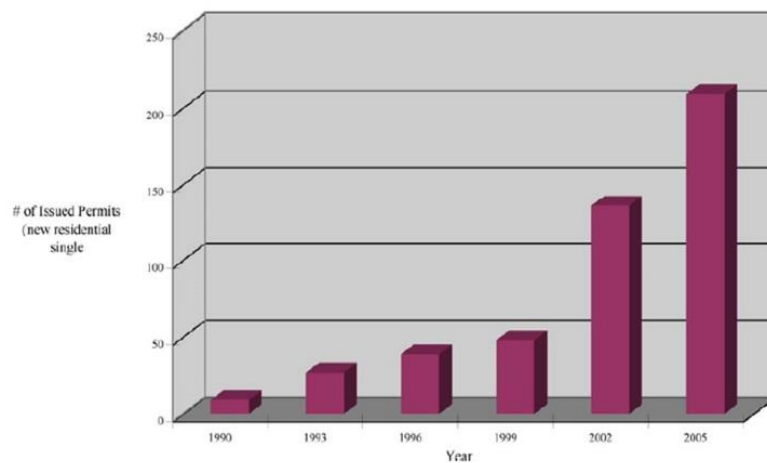


Figure 3. Number of Issued Permits by Year (single family residential and residential remodel)

(Mathon 2014)

and/or what is termed as "white flight". In the renewal/gentrification stage, middle income residents move back for cheaper housing and a convenient central location. These among other reasons increase the housing prices in those neighborhoods (Kirk and Laub, Neighborhood Change and Crime in the Modern Metropolis 2010).

Covington and Taylor 1989 conceptualized gentrifying neighborhoods as those that have experienced a rapid increase in relative house value, educational levels and the percentage of managerial-professional workers. Before gentrification happened, these neighborhoods were considered below average in stability and status. However, these neighborhoods were above average in poverty rate and vacant housing compared to the rest of the city. The stability of a neighborhood depends on the percentage of owner-occupied houses and married-couple households (Covington and Taylor 1989).

Gentrification and Crime

One of the selling points of gentrification is that crime decreases as a result of it. However, the perception of gentrification varies among the races. New white residents tend to see the areas they live in as spotty with "good" and "bad" areas of crime, and with each new area having a new name associated with relatively recent changes. In contrast, the longtime non-white residents of a place tend to see their neighborhoods more holistically, both in terms of crime rates and the names and boundaries of communities (Hwang 2015). This study will examine the validity of that thinking for Austin from 2003-2013. A number of studies into other major cities have looked into in this and have varying results of their research. One line of thinking is that gentrification increases the

opportunity for property crimes because the disruption of social networks and social control processes may cause an increase in crime, especially in property crimes (Covington and Taylor 1989); (Van Wilsem, Wittebrood and De Graaf 2006). A second competing hypothesis is that the gentrification process pushes more crime prone individuals out and therefore reducing the overall crime (McDonald 1986); (O'Sullivan 2005); (Papachristos, et al. 2011). A third hypothesis is that there is a curvilinear relationship between gentrification and crime. Gentrifying neighborhoods will initially experience a short term increase in crime followed by a decrease thereafter (Covington and Taylor 1989); (McDonald 1986); (Papachristos, et al. 2011). This study focuses on how crime is affected by gentrification and seeing which of these three hypotheses will most accurately describe the crime and gentrification relationship in the areas studied.

Thesis Statement and Research Questions

The thesis statement for this study is as follows: Austin has experienced rapid growth leading to gentrification in some areas in the past two decades and these changes have affected the crime rate in the affected areas.

The research questions this study will address are as follows. How has the crime rate changed from 2003 to 2013 for violent, property and non-indexed crimes during the gentrification process for census tracts marked as "gentrified" as compared to ones that were marked as "did not gentrify"? Which types of crimes change with gentrification? Is there spatial variation throughout the study area on how gentrified and non-gentrified tracts crime rate change with crime?

Limitations of the Study

The entire study can be done remotely using information gathered from the public domain. The strength of this approach is that it can be reproduced for other cities and by another researcher for Austin. The weakness of this approach, as brought up by Kreager, Lyons and Hays (2011), is that relying solely on administrative data may lack sufficient detail to capture the gentrification process and utilizing qualitative field surveys to identify gentrifying areas may be a more accurate way to identify gentrification (Kreager, Lyons and Hays 2011).

Another limitation is the relatively large scale of the adopted unit of analysis (census tracts), which prevents analysis at a greater detail. Census blocks would be a better choice since it would be more at a neighborhood level and is available for decennial census data, but is not available for crime data. With crime data no more refined than a census tract, there is no reason to refine the census data any more than the tract level.

Justification of this Study

Almost everyone is in agreement that it is a good idea to help economically depressed areas of the city. The problem comes when actions perceived as improvements in the community do more harm than good. The term "gentrification" has negative connotations associated with it and is almost never used from those who support it. The reason why this study is so important is that so many people are affected by gentrification and crime reduction is one of the selling points for supporting it. For city planners,

improvement projects are one of the major methods to help improve neighborhoods, but can also lead to gentrification. To pay for the projects, investors, new businesses and new residents with much greater buying power than the existing residents are needed to pay for the projects. Otherwise, more tax dollars are used, which is never popular. In the end, existing residents with many going back several generations, can be priced out. So if the selling point of reduced crime is not valid, then other factors will need to justify these projects before more investment is done that could lead to gentrification.

II. BACKGROUND AND LITERATURE REVIEW

Criminology Literature on Gentrification

Criminology theory is used to interpret the relationship between gentrification and crime in many studies and used as a tool for interpreting observations and results. Human ecology and social disorganization perspectives view gentrification as a disruptive force to informal social controls that potentially contribute to increase in crime (Kreager, Lyons and Hays 2011). McDonald (1986) stated that gentrification may increase crime by 1) The number of attractive targets increases 2) increase inequity in the neighborhood and therefore increasing intra-community conflict. 3) Reduction of informal crime controls by breaking the longstanding ties held by previous residents (McDonald 1986). The streams of criminology literature that are drawn on in this study include: Social Disorganization Theory, Routine Activities Theory, Collective Efficacy Theory, Systemic Models, Social Learning Theory, and Relative Deprivation.

Social Disorganization Theory is one of the main areas of criminology that seeks to explain how crime rates change during the gentrification process. A recurring theme in criminology theory is that residential instability and population heterogeneity reduce a community's ability to organize for the common good including crime reduction (Kreager, Lyons and Hays 2011). As social structures undergo a period of flux at the beginning of the gentrifying process, neighborhoods may experience an increase in crime by lessening a community's ability to control crime (Kreager, Lyons and Hays 2011). Sampson and Groves tested Shaw and McKay's 1989 community Social Disorganization Theory on crime by analyzing the data from surveys performed in Great Britain. The

results show that variations between communities in social disorganization affect community structural characteristics and affect the rate of both criminal victimization and criminal offences (Sampson and Groves, *Community Structure and Crime: Testing Social-Disorganization Theory* 1989). Anderson (1990) states that when gentrification stalls, the surrounding community loses its controls of crime. Stalling can be from higher income families seeking better school districts when their children reach school age or when they personally experience crime. This in and out of residents affects the social networking needed to ensure the controls against crime remain in place (Anderson 1990).

Systemic models of social organization explain communities as complex social networks from close friendships to “diffuse associations” (Sampson, Raudenbush and Earls, *Neighborhoods and Violent Crime: A Multilevel Study of Collective Efficacy* 1997). The focus of systemic models is on the structures of relationships among people (social capital) that facilitate action (Coleman 2003). Systemic models point to gentrification as one of reasons why these networks can either be destroyed or rebuilt depending on the stage of the gentrification process as brought up later in the paper. Local friendship networks, control of local teenage peer groups, and the presence of organizational participation make up a communities level of social organization. However, low economic status, ethnic heterogeneity, residential mobility and family disruption lead to social disorganization and an increase in crime rate among other social issues (Shaw and McKay 1931). Atkinson (2000) had findings supporting the systemic model in central London with similar conclusions that extensive gentrification in the areas studied threatens the social networks within the community and argues that there is

a need to rethink the "laissez faire policy" of injecting wealthy households in poorer communities (Atkinson 2000).

In Routine Activities Theory, Cohen and Felson (1979) state that structural changes in routine activities influence direct-contact predatory crime rates when at least one of the three is affected at a convergence of time and space: 1) motivated offenders 2) suitable targets, and 3) absence of capable guardians against a violation. For example, if the number of motivated offenders and suitable targets remain the same and somehow the number of capable guardians goes down, then crime rates will increase (Cohen and Felson 1979).

Collective Efficacy Theory is related to systemic models, but goes more into the psychology of networking among neighbors. It states that the more the social cohesion neighbors have with one another, the less violence a neighborhood will have according to Sampson, Raudenbush and Earls (1997). This is because closely knit neighbors will intervene on behalf of the common good when necessary. Any disruption of this may cause a rise in crime (Sampson, Raudenbush and Earls, *Neighborhoods and Violent Crime: A Multilevel Study of Collective Efficacy* 1997).

In Social Learning Theory, individuals experience social learning through imitation and reinforcement in their environment through favorable and unfavorable outcomes or their behavior with peer affiliations having a major role (Akers 2009). Bursik and Webb (1982) state that social institutions and social networks are greatly affected when a community almost changes completely in a short period of time. Even

the institutions that persevere may resist the inclusion of new residents (Bursik and Webb 1982).

Relative Deprivation Theory states that the greater the relative difference in economic inequality within a community, the greater the crime rate will be. "...income inequality in a metropolis substantially raises its rate of criminal violence" (Blau and Blau 1982).

Gentrification Increases Crime

Covington and Taylor (1989) found a positive relationship between gentrification and violent crime in Baltimore neighborhoods between 1970 and 1980. They found that when a neighborhood rapidly increased in "status" (i.e. greater home values, less poverty, and higher level of education) or "stability" (i.e. more married couples and owner occupied houses) the more they saw an increase in homicides and aggravated assaults. They interpret these findings using a human ecology framework in that the social change caused from gentrification, even if it's in a positive direction, can disrupt the social processes that enable crime control. They found that relative resource deprivation had less of an effect than social control and neighborhood stability (Covington and Taylor 1989).

Continuing with Taylor and Covington's conclusions regarding rapid neighborhood change and an increase in crime (victimization of theft, violence, and vandalism), Van Wilsem, Wittebrood and De Graaf (2006) also linked rapid economic improvement with an increase in crime for neighborhoods in the Netherlands consistent

with the systemic model. The inflow of residents hindered social control negatively affecting cohesion among neighborhood residents (Van Wilsem, Wittebrood and De Graaf 2006).

Gentrification Decreases Crime

McDonald (1986) analyzed crime rates in 14 gentrifying neighborhoods in five major US Cities from 1970 through 1984 and found that gentrifying neighborhoods experienced a reduction in violent crime, but no significant change in property crime. McDonald wrote that gentrification may reduce crime by 1) Displacing the "criminal elements" with less crime prone residents that are more affluent. 2) Increasing neighborhood controls where new residents are better organized to fight crime and 3) Renovations in the gentrifying neighborhood instill greater pride in it and subsequently decreasing criminal opportunities (McDonald 1986).

In their study on gentrification and crime in Seattle, Kreager, Lyons and Hays (2011) point to the negative relationship between gentrification and crime and suggest that the criminal elements in the affected areas were either deterred by an increase in enforcement efforts, an increase in economic and social opportunities got them away from criminal activities or they were displaced as a result of these changes in the neighborhood. The more recent consolidated gentrification of the 1990's may actually reduce crime by undermining criminogenic conditions by displacing poverty along with motivated offenders and increasing formal controls such as increased police protection and building surveillance (Kreager, Lyons and Hays 2011).

Papachristos, et al. (2011) state that the level of gentrification can be attributed to a decrease in violent crime. One way to test this is to pick a common indicator such as the number of coffee shops as a measure of how well an area has "re-organized" itself after undergoing changes due to gentrification. This follows the previous theories of social disorganization that indicate a higher crime rate with social disorganization such as the beginning stages of gentrification or non-gentrification in high crime areas. A study in Chicago between 1991 and 2005 indicated a negative relationship between the number of coffee shops to homicide rates. This study confirmed its hypothesis that as the number of coffee shops increased, the number of homicides decreased accordingly. (Papachristos, et al. 2011).

O'Sullivan (2005) performed a case study regarding crime and gentrification in Portland, Oregon from 1990 to 2000. The study showed that parts of downtown Portland experiencing gentrification had a reduction in crime rate at a faster pace than the rest of the city during the same time period. During this time period, these same areas of gentrification had an increase in educated higher income households and a displacement of low income and black households. The tracts with the largest reductions in crime also had the largest rate of increase of housing prices. Interestingly, these same tracts also had some of the highest crime rates in 1992 (O'Sullivan 2005).

Gentrification and Crime Have a Curvilinear Relationship

The causal relationship between crime and gentrification is not simple and not linear. Feedback from each are mutually reinforced (Papachristos, et al. 2011). Covington

and Taylor's 1989 report did acknowledge that the study was done in Baltimore from 1970 to 1980 was sporadic on a house by house basis and did not cover vast areas opening the possibility that it is an unknown whether the disruption of systemic organization was a factor in larger areas of gentrification as seen in the 1990's (Covington and Taylor 1989). Gentrified neighborhoods may experience a short term rise in crime because of the social instability associated with it, but decline in the long term as social stability takes over (McDonald 1986). As Covington and Taylor (1989) point out, after a certain "tipping point", crime may decline as a result of added controls in the affected population (Covington and Taylor 1989) .

Kreager, Lyons and Hays (2011) states that the collective efficacy of a neighborhood will initially be negatively affected by gentrification. As more of a neighborhood is gentrified though, the new neighbors will redevelop cohesive relationships. The relationship between gentrification and crime may change depending on the stage of the gentrification process. When gentrification is "spotty" as it was in the 1970's and 1980's, social disorganization theory more accurately predicts the increase in crime. Much of the gentrification that occurred in the 1990's was more on a grand scale and funded by large corporations rather than just individuals and transforming entire city blocks rather than one house at a time. This large scale transformation created a greater population turnover and enhanced formal controls such as greater police presence, improved security systems and large groups of closely knit new residents creating their own social networks (Kreager, Lyons and Hays 2011).

Types of Crimes Affected

Neighborhood stability was found to have more of a positive effect on violent crime than neighborhood status making them conclude that informal social networks can be disrupted by gentrification increasing group conflict and thereby ripening conditions for crime (Covington and Taylor 1989). However, gentrification can create more lucrative opportunities for particular types of crimes, especially property crimes (Covington and Taylor 1989). Kreager, Lyons and Hays (2011) found a negative relationship between gentrification and violent crime (Kreager, Lyons and Hays 2011). McDonald (1986) found that personal crime (i.e. homicide, rape, robbery, and aggravated assault) decreased and property crime had no significant change in a study of 14 gentrifying neighborhoods in five major U.S. cities from 1970 to 1984 (McDonald 1986).

Gentrification Depends on Crime

While there is debate on the positive and negative relationship between gentrification and crime, there appears to be some evidence, according to Papachristos, et al (2011), that gentrification is at least somewhat dependent on crime rates. Some suggest individuals and corporations will only invest in high risk areas of town only if they see the crime rate trending downward leading to the belief that gentrification occurs after crime rates have gone down. Once gentrification starts, the crime rate may further decrease (Papachristos, et al. 2011). Clay and Taub (1979) found that the middle class will consider moving to lower income urban centers if they perceive that the crime rate has at least stabilized (Clay 1979).

Ellen and O'Regan 2008 state that prior to the 1990's, high crime rates were a contributing factor for population loss in cities. Therefore, a the decline in crime in the 1990's and beyond could help contribute to repopulating the same areas affected by crime such as the poorer sections of the inner city that originally lost the most people (Ellen and O'Regan 2008). Kirk and Laub (2010) studied how the crime rate affects property values in the 1990's. They state that a decline in crime may encourage residents to invest in neighborhoods where crime has decreased (Kirk and Laub, Neighborhood Change and Crime in the Modern Metropolis 2010). According to O'Sullivan (2005), future gentrifiers have more to lose in high crime neighborhoods leading them to go through an informal cost benefit analysis to determine whether the shorter commute among other benefits outweigh the higher risk of crime. If indeed this is the case, then gentrification would more likely occur (O'Sullivan 2005).

Schwartz, Susin, and Voicu (2003) examined how the decline in crime in the New York City in the 1990's helped drive up property values by 17.5 percent from 1994 to 1998 and concluded that the decline in crime contributed to about one third (about 6 percent of the total) of this increase. They also found that violent crime affected property values more than property crime (Schwartz, Susin and Voicu 2003).

Crime Displacement as a Result of Gentrification

Studying gentrification and crime is not complete without knowing how crime is displaced during and after the gentrification process, yet investigations of the displacement of crime from gentrification are lacking (Kirk and Laub 2010). One of the

few studies of crime displacement was done by Curtis (1998) showing how crime was displaced from Williamsburg to Bushwick within Brooklyn from 1987-1997.

Gentrification in lower Manhattan made the neighborhoods in the outer boroughs, like Williamsburg and Bushwick in Brooklyn, attractive to many professionals and artists.

However, Williamsburg was better positioned than Bushwick geographically making it gentrify at a much higher rate. As a result, many drug dealers and other criminals moved over to Bushwick and increased its crime rate accordingly (R. Curtis 1998).

Literature Review Summary

Research on gentrification and crime have found diverging opinions on the whether the relationship is positive or negative between the two with empirical evidence supporting both sides (Papachristos, et al. 2011). Literature on the relationship between gentrification and crime is plentiful, but the number of actual studies are lacking.

Therefore, most of the literature is based on criminology theory with some studies of gentrification and crime in major cities along the way.

III. RESEARCH DESIGN AND METHODOLOGY

Criteria for Choosing Gentrified and Non-Gentrified Tracts

The criteria for what is considered gentrified in this study was adopted from the "Governing" website's criteria. The "Governing" website shows the affected census tracts on an interactive map (Governing 2015). This web page uses the criteria for gentrification based on Freeman's 2005 *Urban Affairs Review* article on the topic (Freeman, 2005). It divides tracts into three categories for a period of analysis: "gentrified", "did not gentrify", and "not-eligible". The map considered in this study is the "Austin Gentrification Map: 2000 Census - Present". Figure 4 shows the screen shot of the map from the web page. " In parts of the analysis in this study, "gentrified"="G" and "did not gentrify"="DNG". Tracts labeled as "not eligible"="NE" were not considered in this study as discussed later.

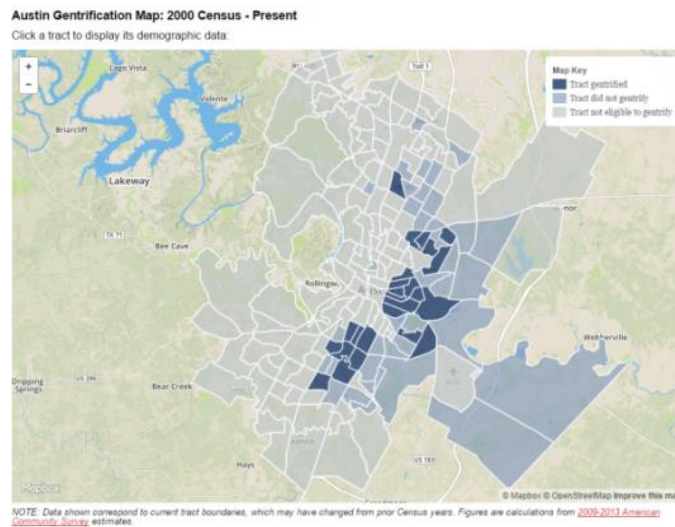


Figure 4. Screen shot from governing.com website
(Governing 2015)

The criteria by governing.com for a tract to be considered "gentrified" are done by two tests with test one being the criteria for the tract at the beginning of the period of consideration and test two done at the end of the period of consideration. The period of consideration for this study is ten years. The first test has three criteria: 1) At the beginning of the period of study, a tract will need a population of at least 500 people, 2) The tract's median household income needs to be in the bottom 40th percentile for the metro area it's located in, and 3) The tracts median home value will need to be in the bottom 40th percentile for the metro area it's located in. The second test after the period of study analyzes the change in the following three areas: 1) the percentage of residents with a bachelor's degree age 25 and older will need to be in the top third percentile for the metro area, 2) the median home value will have increased for a tract when adjusted for inflation, and 3) the percentage increase of inflation adjusted median home value is in the top third percentile for the metro area. The criteria for test 1 and test 2 help ensure that the tracts labeled as "gentrified" in this study are not counted based on normal increases in education, property values and family income and to ensure that only tracts that were considered in the study were under average for all three criteria at the beginning of the study. From the Gentrification Report Methodology section in the Governing website, tracts labeled as "not eligible" either had missing data or did not meet any one of the measures in test 1 or test 2 stated above. For this reason, "not eligible" tracts were left out of the study.

Crime Data

The crime data used in this study are for census tracts from 2003 to 2013. The crime data is from the “crime” section of the austintexas.gov website (austintexas.gov 2015). The types of crime considered in this study are termed "indexed" and "non-indexed" crime. Indexed crime is: murder, rape, robbery, aggravated assault, burglary, theft, motor vehicle theft, and arson. The total indexed crime in the report is not a simple sum of all the individual indexed crimes listed. For example, combined index column also includes "criminal negligent homicide" and "involuntary manslaughter" that is not included in the "murder" column. "Non-Indexed" crime includes all other crimes and including "simple" assault, forgery, weapons violations, drugs, criminal mischief, public intoxication, disorderly conduct, and various city ordinance violations among many others.

In the statistical analysis, all the crimes listed above were combined and categorized into three groups: violent crime, non-violent crime and non-indexed crime. Violent crime is the combination of robbery and aggravated assault. Non-violent crime is the combination of burglary, motor-vehicle theft, and theft. Non-Indexed retains the original data. These various combinations are the reason why the total number of observations varies for each crime category. Rape, arson and murder were omitted for lack of data. In this study, the term "three types of crime" refer to the combination of property, violent and non-indexed crime. The disaggregated results showing all the data on all of the crimes can be seen in the Table 1 in the "Data Analysis/Results" section.

Preparing the Crime Data for Analysis and Defining the Study Area

The available data are on the austintexas.gov web site in a pdf format from 2003 to 2013 by both census tract and zip code. For comparative reasons tract level crime data was chosen since it was going to be analyzed with tract level information on gentrification. Both the 2003 and the 2013 pdf files were then converted to Excel files for immediate analysis and for joining shape file tables for geographic information system (GIS) analysis. This involved modifying the tract level fields and crime fields to be text and numeric fields respectively. The crime data for both 2003 and 2013 were then normalized per 1000 people based on decennial data from the United States Census Bureau for 2000 and 2010 respectively. This normalized value will be called the "crime rate" and will be used for the remainder of the study. The shape files for the 2000 census tracts and all of decennial data are from the United State Census Bureau (United States Census Bureau 2015). The major roads and Austin shape files are from Capital Area Council of Governments (Capital Area Council of Governments 2015). ArcMap 10.2.2 was the GIS software used to create maps and SAS JMP 12 was used for statistical analysis.

Census tracts are updated every ten years to maintain roughly the same population every decade making it necessary to split the census tracts up as the population increases. The 2003 crime data goes by the 2000 census tracts and the 2013 data goes by the 2010 census tracts. With the population growth between 2000 and 2010, many census tracts from 2000 were split creating an issue with comparative analysis between 2000 and 2010. Because of this, it was decided that the 2000 census tracts will be used for both the 2003 and 2013 crime data to fully compare the tracts. This involved aggregating several 2010

tracts back to their 2000 counterparts. This also brought up issues when a mixture of gentrified and non-gentrified tracts from 2013 were combined into one 2003 tract. For example, the year 2000 tract of 23.11 was split into 23.17 and 23.18 in 2010. Tract 23.17 was labeled as "Did Not Gentrify" and 23.18 was labeled as "Gentrified". As a result, tract 23.11 was eliminated from the study. The crime rate was then matched for the year 2000 census tracts for gentrified, non-gentrified and the rest of the city tracts. The tracts used for this study included all of the tracts that were labeled as "Tract Gentrified" and "Tract Did Not Gentry" as shown in Figure 5. In all, 24 tracts were selected from each category.

Testing the Data

Once the change in crime per 1000 people are gathered for property, violent and non-indexed crime, a goodness-of-fit test will be performed before data analysis is done to determine whether the data is considered normal enough in each of these types of crime. A parametric statistical analysis will be done if the data is normal and a nonparametric analysis will be done if the data is determined to be non-normal. Once the normality is determined, a statistical analysis will be performed to see whether the medians of the change in crime for property, violent and non-indexed crimes separately for "gentrified" and "did not gentrify" census tracts are statistically different from one another.

Spatial Variation of Crime and Gentrification

In studying the spatial variation for "gentrified" and "did not gentrify" tracts for the rate of change in crime for all three types of crime, Austin was divided into six parts: east central Austin, northeast central Austin, southeast Austin, north east Austin, far east Austin, north Austin. Tracts outside the study area were aggregated and are labeled as "rest of Austin".

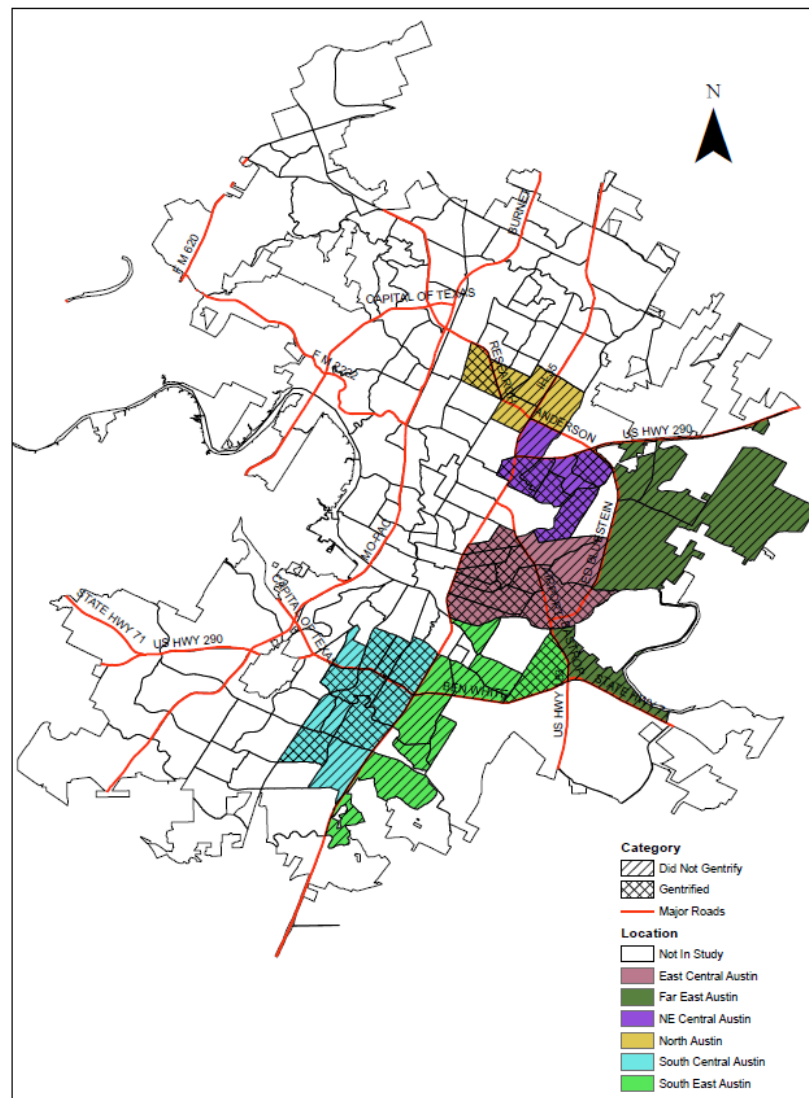


Figure 5. 2000 Census Tracts in Study with Surrounding Tracts

IV. DATA ANALYSIS / RESULTS

Descriptive Analysis

Before the data are normalized and changes in crime are shown, it is worth showing the number of crimes that are behind the analyses. Table 1 compares the types of crime between 2003 and 2013. Murder, Rape and Arson combined make up less than one percent of the total crime in 2003 or 2013 and left out of the study. Theft makes up most of the property crime category.

Table 1. Percentage of Indexed Crime per Type of Crime Plus total Indexed Crime (2003 and 2013)-Grey Rows Used In Study

Aggregated Crime	Crime	2003	Percent of Indexed Crime	2013	Percent of Indexed Crime
Violent Crime	Robbery	1211	2.90%	805	1.80%
	Aggravated Assault	1415	3.30%	1969	4.30%
Property Crime	Burglary	6323	14.90%	6399	14.10%
	Theft	30172	71.10%	33505	73.80%
	Motor-Vehicle Theft	2934	6.90%	2380	5.20%
Crime Omitted From Study for Lack of Data	Murder	32	0.10%	30	0.10%
	Rape	225	0.50%	252	0.60%
	Arson	125	0.30%	86	0.20%
Sum of Indexed Crime		42437		45426	
Non-Indexed Crime	Non-Indexed Crime	80,916		89,385	

Figure 6 compares how the average crime rates stack up to one another by year and gentrification. On average, the rest of Austin has about a third of the violent Crime rate and half the non-indexed crime rate as the study area. This makes sense because the original tracts were selected because of the criteria on gentrification would put these tracts at a higher risk for violent and non-indexed crime. As mentioned in the literature review, property crime is just as prevalent in areas that are not economically challenged and is shown in several studies and to be less dependent on economic and demographic changes in a tract.

The descriptive statistics shown in Figure 7 show that crime decreased in the study areas as well as the rest of Austin. On average, the non-gentrified areas had a greater reduction in crime. As other studies have pointed out, tracts with higher crime rates at the beginning of the study tend to experience a greater percentage reduction by the end of the study period. One can see this when comparing Figure 6 and Figure 7. For example, Figure 6 shows the average property crime per 1000 people for non-gentrified tracts in 2003 and 2013 was 66.9 and 47.2 respectively.

The average property crime per 1000 people for the "rest of the city" tracts in 2003 and 2013 was 50.5 and 44.5 respectively. Even though the crime has reduced at a higher rate in the gentrified and non-gentrified areas of Austin than the rest of the city, the rest of the city still has less crime in all categories. Figure 7 shows that in property crime and non-indexed crime that non-gentrified tracts have a greater reduction in crime than gentrified tracts. This is not the case for violent crime where gentrified tracts have a greater reduction in crime. Whether the medians of the non-gentrified and gentrified are statistically different will be shown later in the analysis.

The maps in Figures 8, 9 and 10 show in visual detail how the crime rate has changed from 2003 to 2013 for all of Austin as compared with the gentrified and non-gentrified tracts. These maps clearly show that the areas with the least change in crime are typically west of I-35. The areas with the greatest reduction in property crime are the south east sections of Austin. The largest increase in violent crimes occurred in the East Riverside and the Ohlen/Research (183) corridors. Violent crime saw the greatest reduction in the heavily gentrified east central Austin area near East 11th and I-35.

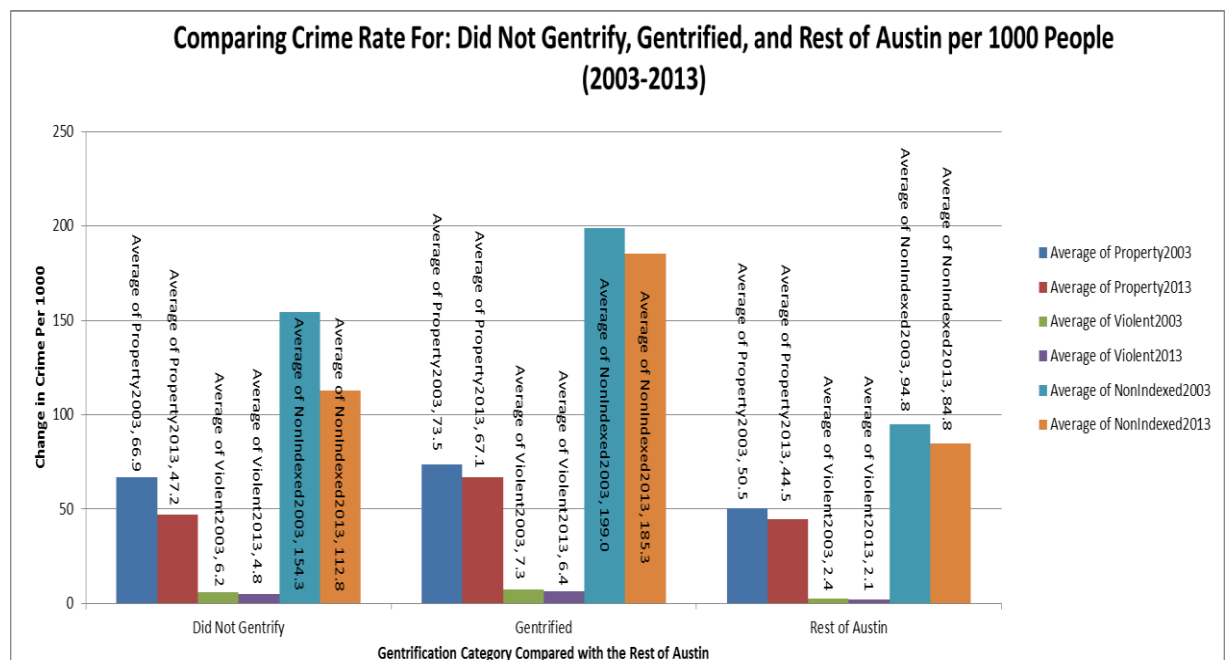


Figure 6. Comparing Crime Rate For: Did Not Gentrify, Gentrified, and Rest of Austin per 1000 People (2003-2013)

Descriptive Statistics for Change in Property, Violent, and Non-Indexed Crime per 1000 People for Austin, Texas (2003-2013)

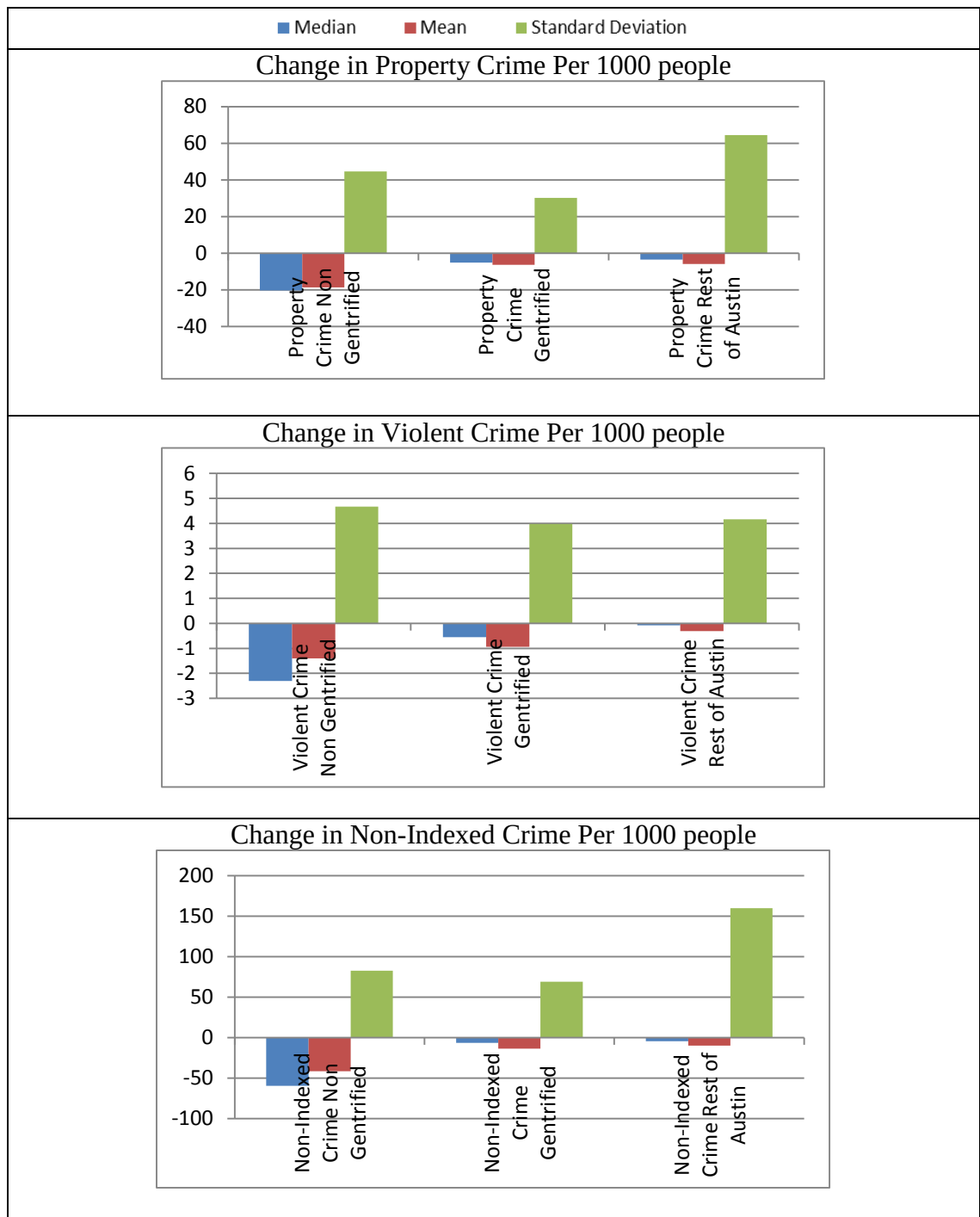


Figure 7. Change in Crime Per 1000 People

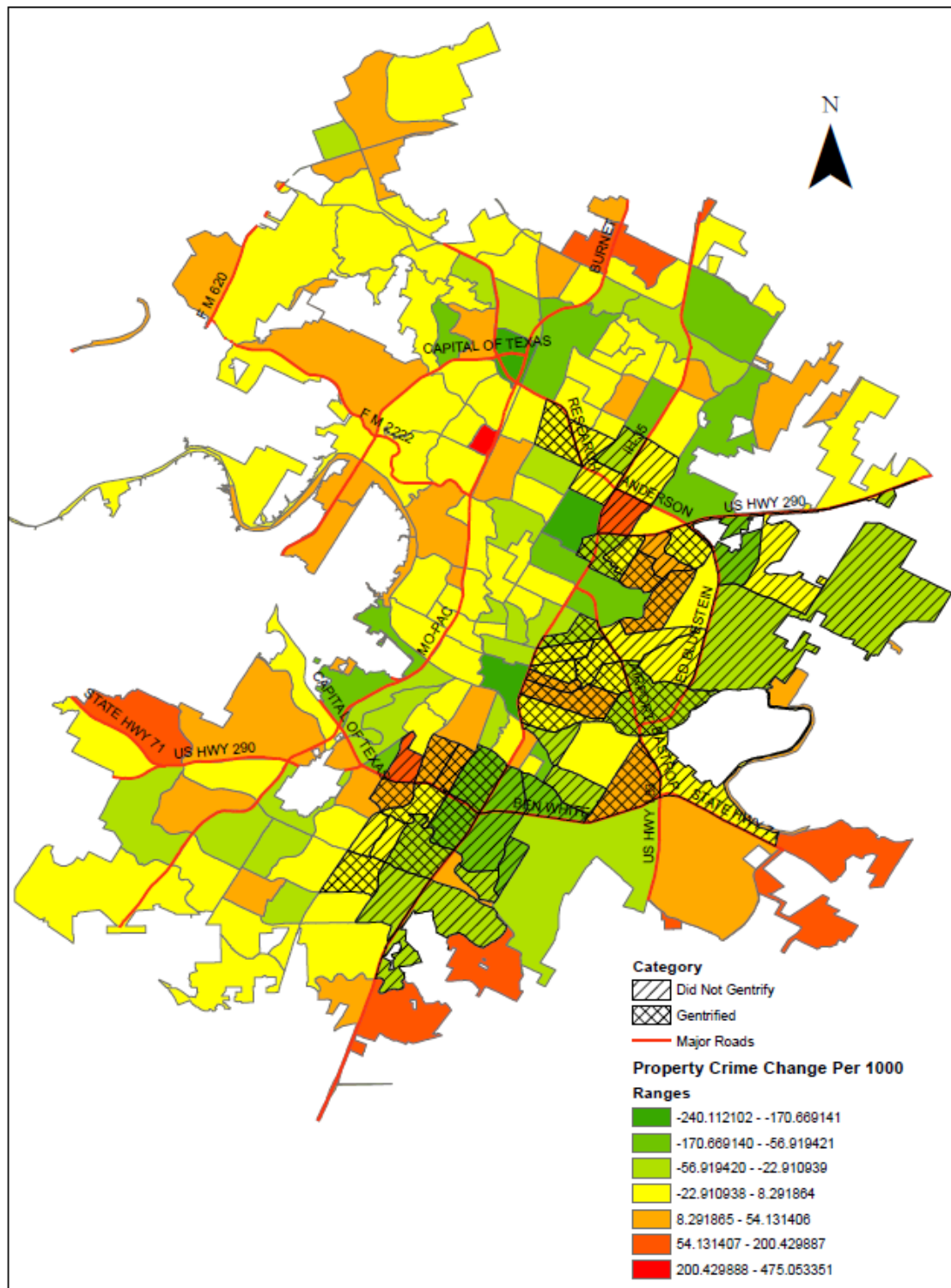


Figure 8. Property Crime Change 2003-2013 per 1000, Austin, Texas

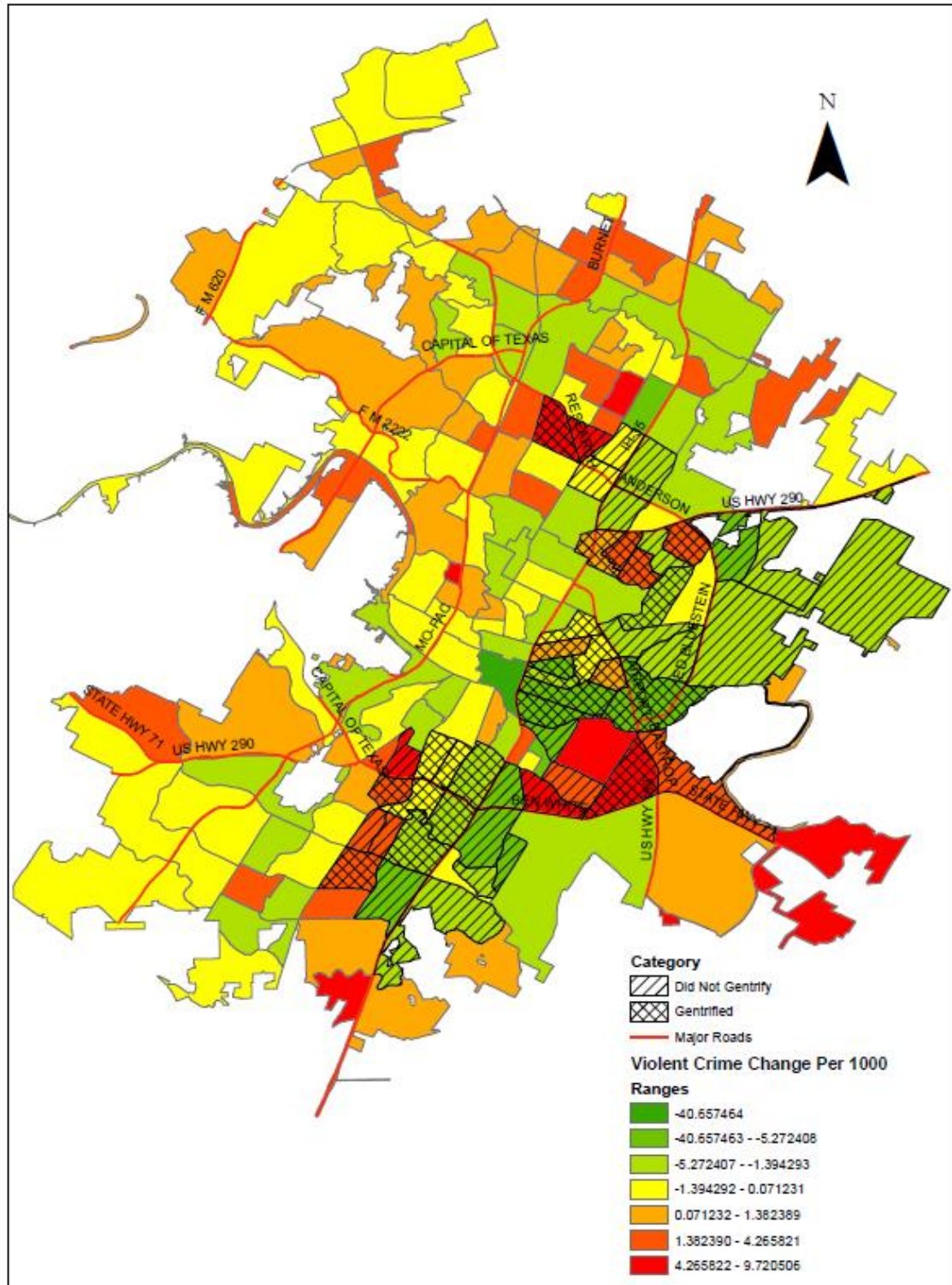


Figure 9. Violent Crime Change 2003-2013 per 1000, Austin, Texas

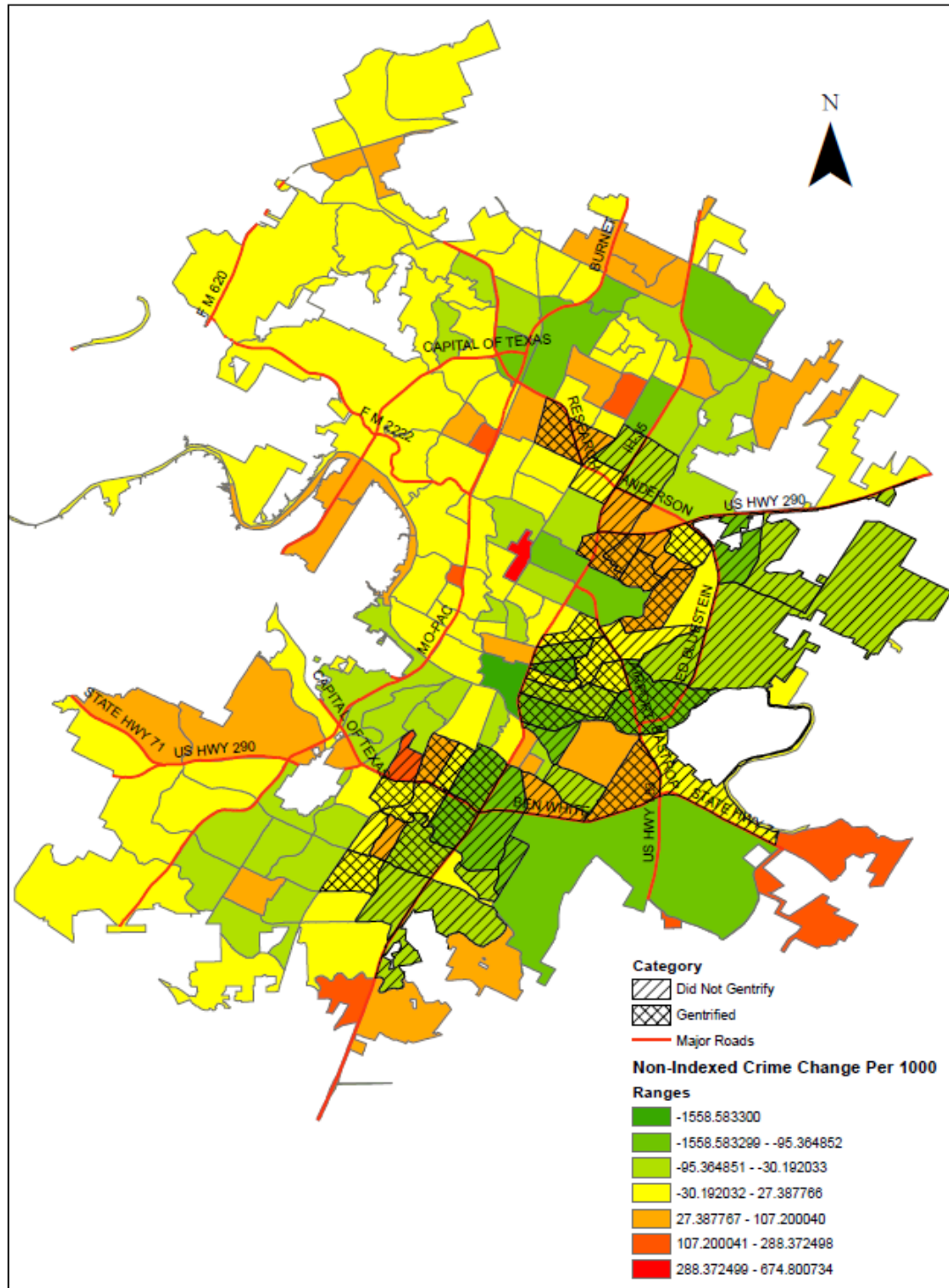


Figure 10. Non-Indexed Crime Change 2003-2013 per 1000, Austin, Texas

Statistical Analysis

As shown in Figure 7 and Figure 11(a), the change in property crime per 1000 box plot indicates that gentrified tracts have a higher median value than tracts that did not gentrify. In Figure 7, Figure 12(a) and Figure 13(a), the change in violent and non-indexed crime per 1000 people box plots indicate that gentrified tracts have a lower median value than tracts that did not gentrify.

In all three types of crime, "did not gentrify" tracts data have more of a spread and has a higher standard deviation than for the "gentrified" tracts. For the change in property crime data, the Shapiro-Wilk W Goodness-of-Fit Test results (Figure 11(c)) along with the visual confirmation of the normal quantile plot (Figure 11(d)) indicate that the data are not normal, so a non-parametric test will be used since the density distribution curves do show that the non-gentrified and gentrified curves are very similar. However, the same tests show that the change in violent and non-indexed crime data is normal based on a level of significance of 0.05, but is not normal based on a level of significance of 0.10. For the sake of consistency with the property crime data, a non-parametric test will be used for both violent and non-indexed crime change since the density distribution curves show that the non-gentrified and gentrified curves are also very similar.

Mann-Whitney/Wilcoxon non-parametric tests for equality of center on all three types of crime, comparing gentrified and non-gentrified tracts from 2003 to 2013 (Table 2), show that we cannot reject the null hypothesis and cannot rule out that both sets of data come from the same population for a level of significance of 0.05.

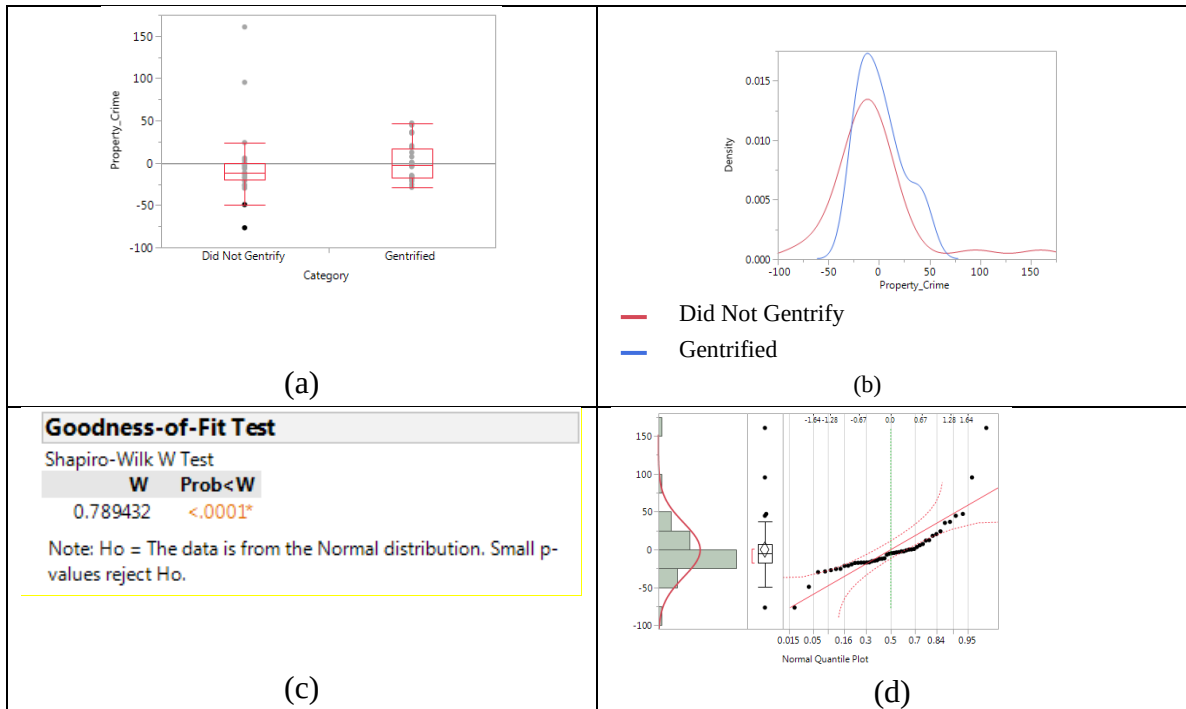


Figure 11. Change in Property Crime Distribution Analysis

Table 2. Change in Property Crime Mann-Whitney-Wilcoxon Analysis

Mann-Whitney Test Parameters	Did Not Gentrify	Gentrified
Sum of Ranks	502	674
U-Test	374	202
Count	24	24
Null Hypothesis	There is no difference between the change in property crime for census tracts that "did not gentrify" from those that "gentrified" from 2003 to 2013 (i.e. the sum of ranks for "did not gentrify" is no different than the sum of ranks that "gentrified").	
Alpha	0.05	
Critical Value	192	
Lowest U-Value Comparison	202 > 192	
Conclusion	Failed to Reject the Null Hypothesis, though would be significant for alpha of 0.1.	

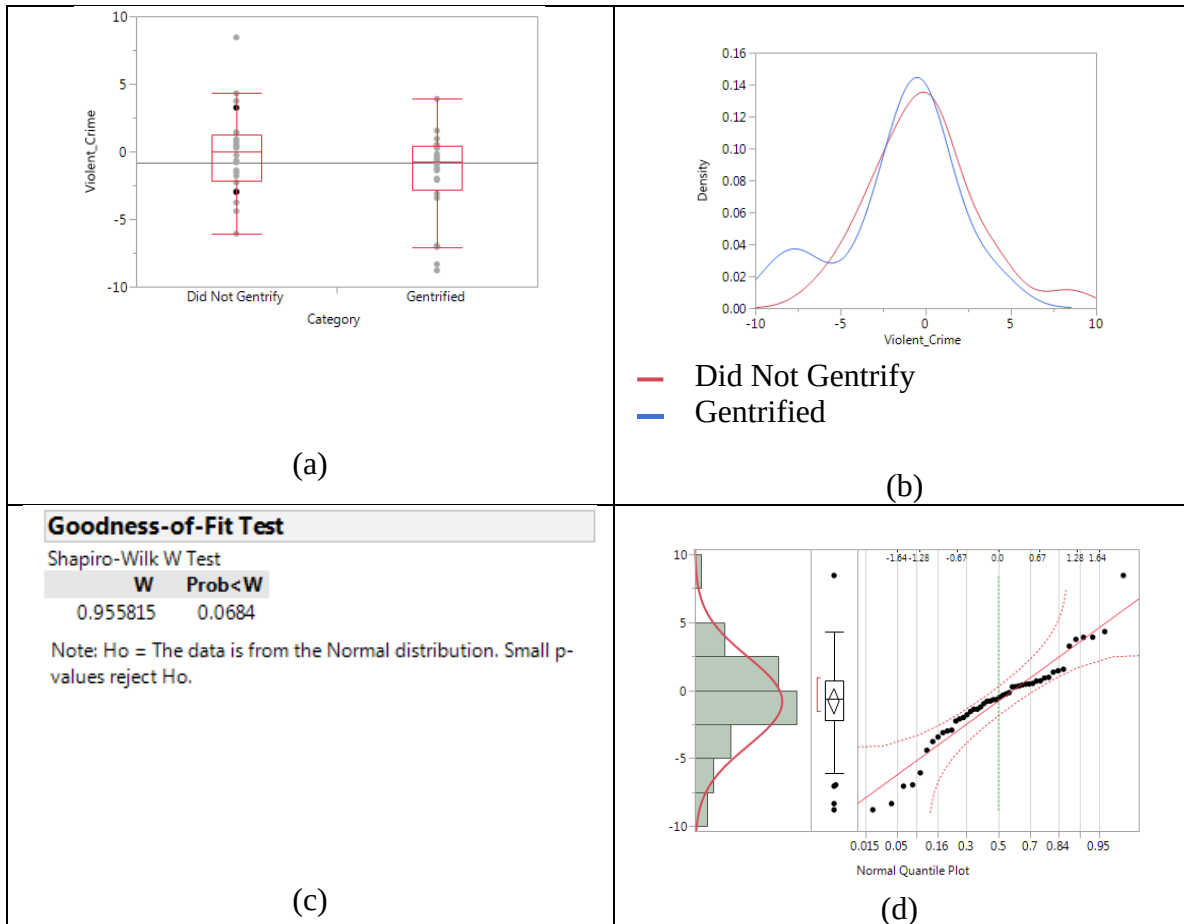


Figure 12. Change in Violent Crime Distribution Analysis

Table 3. Change in Violent Crime Mann-Whitney-Wilcoxon Analysis

Mann-Whitney Test Parameters	Did Not Gentrify	Gentrified
Sum of Ranks	557	619
U-Test	319	257
Count	24	24
Null Hypothesis	There is no difference between the change in violent crime for census tracts that "did not gentrify" from those that "gentrified" from 2003 to 2013 (i.e. the sum of ranks for "did not gentrify" is no different than the sum of ranks that "gentrified").	
Critical Value	192	
Lowest U-Value Comparison	257>192	
Conclusion	Failed to Reject the Null Hypothesis	

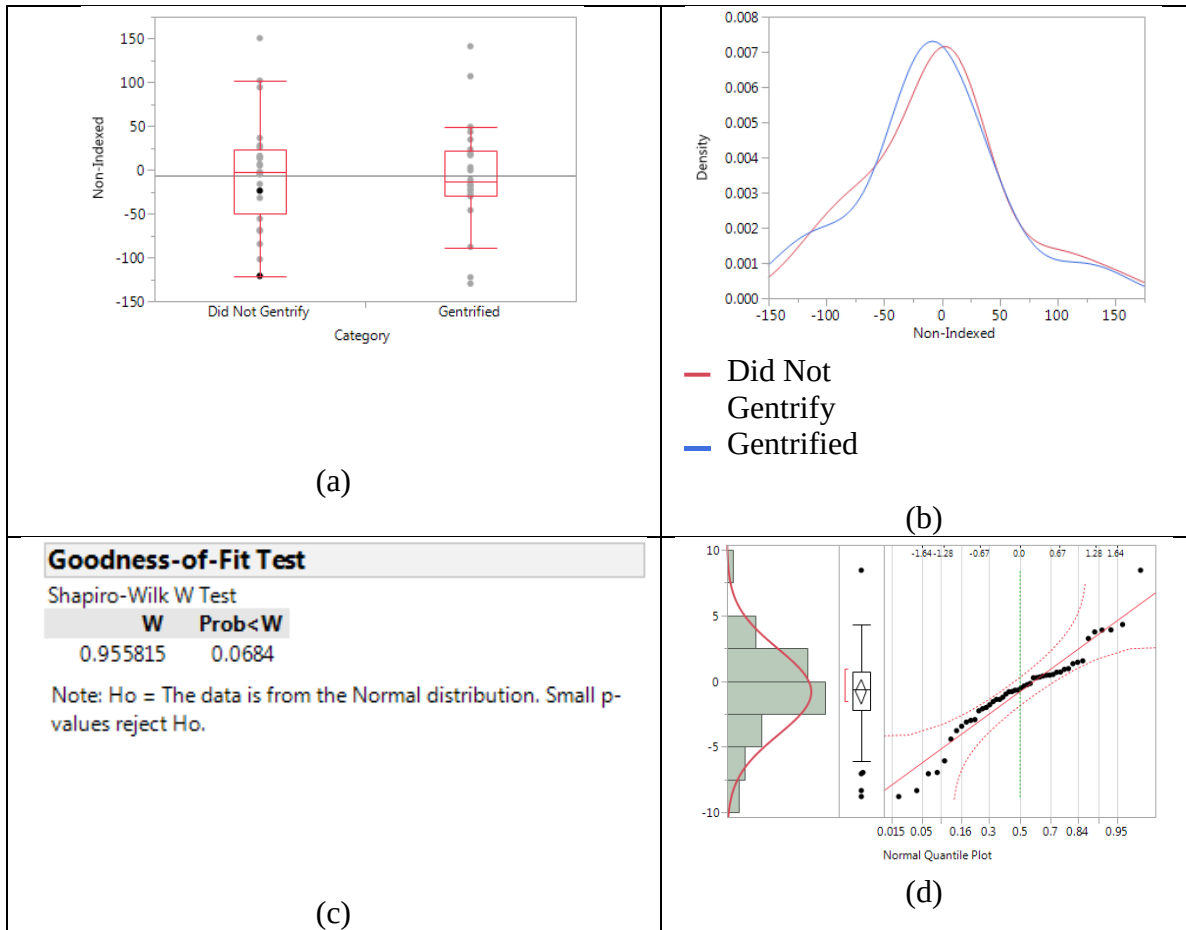


Figure 13. Change in Non-Indexed Crime Distribution Analysis

Table 4. Change in Non-Indexed Crime Mann-Whitney-Wilcoxon Analysis

Mann-Whitney Test Parameters	Did Not Gentrify	Gentrified
Sum of Ranks	605	571
U-Test	271	305
Count	24	24
Null Hypothesis	There is no difference between the change in non-indexed crime for census tracts that "did not gentrify" from those that "gentrified" from 2003 to 2013 (i.e. the sum of ranks for "did not gentrify" is no different than the sum of ranks that "gentrified").	
Critical Value	192	
Lowest U-Value Comparison	271 > 192	
Conclusion	Failed to Reject the Null Hypothesis	

Spatial Variation of Crime and Gentrification

The spatial variation for the rate of change in crime for property, violent and non-indexed crime in the city is shown in Figure 14. The averages of the gentrified and non-gentrified tracts for each defined part of the city as shown in Figure 5 are compared here. Note that the "Far East Austin" and the "Rest of Austin" sections have no gentrified tracts to compare to.

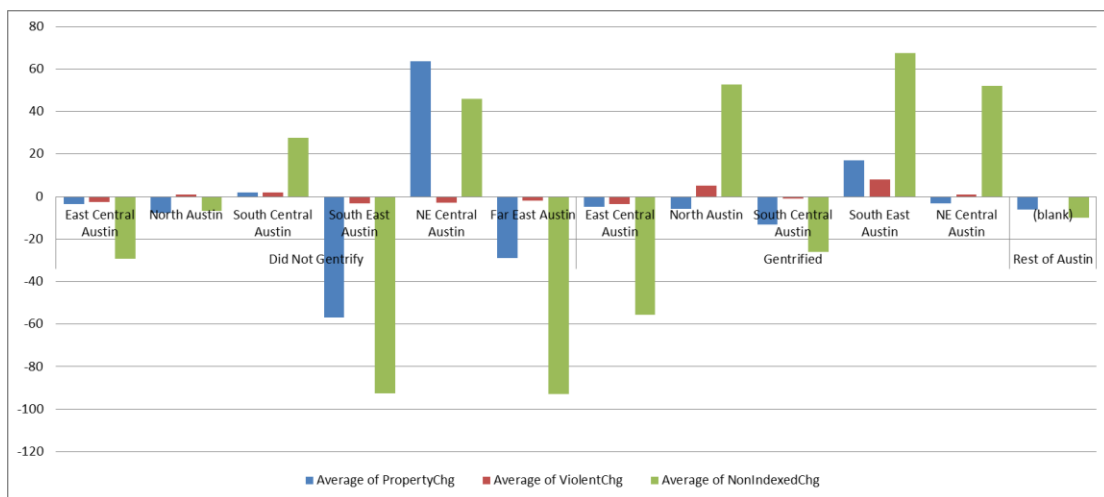


Figure 14. Spatial Variation for: Did Not Gentrify, Gentrified, and Rest of Austin per 1000 People (2003-2013)

V. DISCUSSION

Previous studies into the relationship between gentrification and crime have had mixed results with crime depending on gentrification as having a positive relationship, negative relationship, curvilinear relationship or even showing that gentrification depends on crime.

This study shows that in Austin, Texas from 2003 to 2013 that there was no statistical difference between the gentrified and non-gentrified crime change for property, violent and non-indexed crime for a level of significance of 0.05. However, property crime was the closest to having a statistical difference between gentrified and non-gentrified crime if a level of significance of 0.10 is used. This means that that gentrified tracts had the most significant change of all the crimes with a positive relationship between gentrification and change in property crime for the 0.10 level of significance. This counters the perception that gentrified tracts are somehow safer than non-gentrified tracts as mentioned by Hwang 2015.

Even though the null hypothesis is the same in all three crime types between the gentrified and non-gentrified tracts, the perception was that there was going to be a significant difference based on previous studies by McDonald (1986), Papachristos, et al. (2011), and Kreagor, Lyons and Hays (2011). The results of this study did not support that.

A surprising result happened in the spatial variation of crime and gentrification though. The "south east Austin" section shows that all three crime rates increased for the average of the gentrified tracts and decreased for the average of the "did not gentrify"

tracts. This supports Covington and Taylor (1989) and Van Wilsem, Wittebrood and De Graaf (2006) work showing positive relationship between gentrification and crime.

VI. CONCLUSION

This study found no statistical relationship between crime and gentrification for property, violent and non-indexed crime in this study. This study did answer all three research questions addressed earlier.

The first research question in how the crime rate has changed from 2003 to 2013 for violent, property and non-indexed crimes during the gentrification process for census tracts marked as "gentrified" as compared to ones that were marked as "did not gentrify", was answered indicating no statistical difference for all three crimes types. The second research question regarding what types of crime vary with gentrification was also answered in the study indicating that property crime had the strongest relationship (positive) with gentrification. The spatial variation analysis showed wide differences among different parts of town to answer the third research question of the study.

More questions have come up from this study that might be worthwhile to pursue in further studies. Would a change in criteria for gentrification affect the relationship between crime change and gentrification for Austin? Is there any evidence that crime has shifted from one area to another? How/why did "south east Austin" as identified in Figure 5 have such a dramatic decrease in crime independent of gentrification?

A suggestion for further study is a more detailed examination of exactly who is moving out during the decline and back in during gentrification will help with the analysis of the impact of these changes. It's quite possible that the type of people who moved out aren't necessarily the ones that are moving back in. Another suggestion is

finding a way to measure gentrification or a change by performing a remote sensing study showing the difference between decades for a city.

Possible implications to the results of this study on a personal level could be a change in perception for people who view the gentrified areas as safe zones to the "bad areas" near them as discussed in Hwang 2015. One of the biggest questions and challenges for local governments and investors is "Does the amount of money going into low income neighborhoods translate into positive changes for everyone involved?" One can look at the building permit study of Mathon 2014 and see how the amount of money spent per census tract on building permits relate to crime, property values, household income, and racial percentages among other factors. This kind of analysis could be helpful to quantify the level of gentrification. The results from these studies may give local governments and investors better tools for their decision process on where and how to spend tax dollars and investment money respectively.

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